

A High-Fidelity Microsimulation API for V2X Exploration

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The V2X evolution

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Intelligent
Vehicle Highway
Systems
(IVHS)

Vehicle
Infrastructure
Integration
(VII)

IntelliDrive

Connected
Vehicles
(CV)

Vehicle-to-
Everything
(V2X)



Why V2X still matters

The need for a V2X-aware
microsimulator API
The TransModeler V2X API
Demo

Then two things happened

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Exploration

- ▶ **AI perception became practical at the roadside.**

Cameras + AI: reconstruct trajectories, estimate conflict probability — even when every vehicle is unconnected.

- ▶ **Vision-first vehicles bet on reading the road unaided.**

Cameras, neural networks, onboard compute, fleet learning, over-the-air updates — moving faster than public infrastructure.

Why does Vehicle-to-Everything still matter?

V2X: shareable, trusted, cooperative

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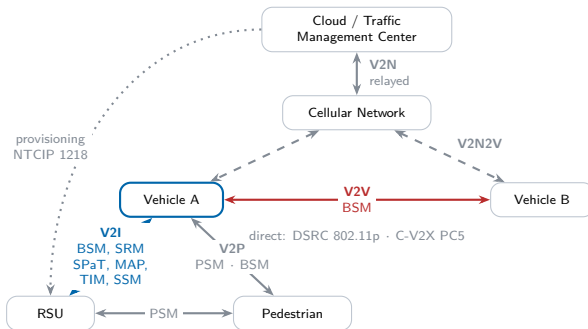
Shareable — one common format, a *stranger* can act on it.

Trusted — *never encrypted*, digitally **signed**, the receiver can verify *origin* and *integrity*.

Cooperative — **direct** (V2V, V2I, V2P) or **network-mediated** (V2N, V2N2V): the *X* is *everything*.

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V2X-aware Simulator

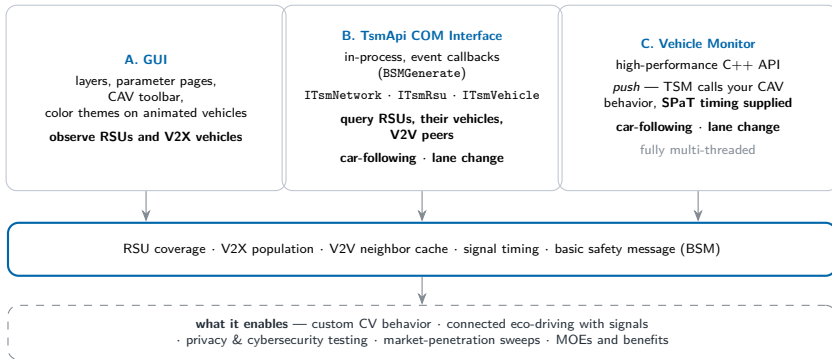
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- 1 **Semantics** — the same concepts real CAV systems use: BSM, SPaT, MAP, PSM, SRM/SSM.
- 2 **Syntax** — SAE J2735 shapes, so a prototype *ports* .
- 3 **RSU emulation** — coverage, placement, malfunction, range variability.
- 4 **Privacy & security** — pseudonym change, silent period, message spoofing — *testable through the API*.
- 5 **User-definable CAV behavior** — the driver model is yours, via API.

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Three surfaces, one simulator

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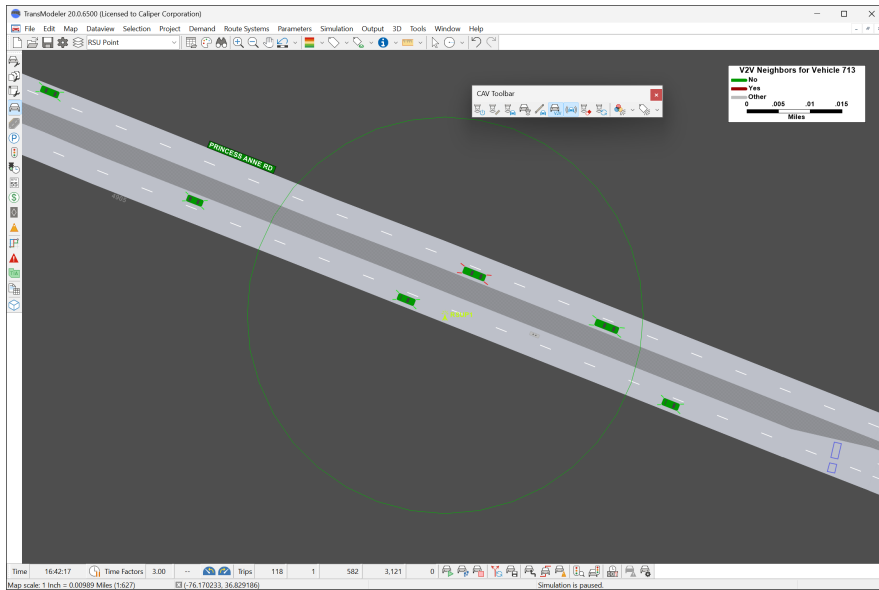
A to *observe* it. **B** and **C** both *redefine the driver* — **B** is easier to reach (C#, any COM client), **C** is what scales.

In practice a plugin uses **both**: **C** for the per-vehicle loop, **B** for the infrastructure it cannot reach — controllers, phases, RSU objects.

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B. TsmApi COM Interface

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ITsmNetwork e.g.

```
— RsuCount
— RsuAt(i)
— RsuByRsuID(bstr)
```

ITsmRsu e.g.

```
— RSUID · Type · Active
— Node(i) · Link(i)
— GetVehicles()
```

ITsmVehicle e.g.

```
— RSU · GetRSUs()
— GetV2VNeighbors(n)
— BSM · GenerateBSM()
— SignalTimeOffMin/Max
```

_IVehicleEvents [source] e.g.

```
— BSMGenerate(...)
— CarFollowingAcceleration(...)
— LaneChange(...)
```

- ▶ **Find RSUs** — by index, numeric ID, or NTCIP string ID.
- ▶ **RSU coverage** — the nodes and the *range in links* it reaches, and the V2X vehicles *currently in range*.
- ▶ **Switch an RSU** — Active is read/write, so a unit can fail mid-run.
- ▶ **Vehicle connectivity** — which RSUs cover this vehicle, which V2V peers are within radio range.
- ▶ **BSM** — read it, overwrite it, or force generation.
- ▶ **Signal timing** — the *current* state of the next signal ahead, with *earliest* and *latest* time it changes. SPaT semantics; **not** a SPaT message.
- ▶ **Callbacks** — BSMGenerate hands you the message **before anyone else reads it**; car-following and lane change let you replace the driver.

C. Vehicle Monitor — the interface

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`CUserVehicleMonitor` *you derive*

```
— RegisterVehicleMonitor()
— AttachVehicle(id, prop, flags)
— StartSimulation / Advance
```

`IUserVehicle` *one per vehicle*

```
— Departure / Arrival
— Update(t, SVehicleBasicState)
— NextSignal(t, SVehicleNextSignal)
— CalculateCarFollowingAccRate(...)
— LaneChange(t, dir, *mandatory)
```

`VM_* flags` *opt in, per vehicle*

```
— VM_CF_SUBJECT · VM_CF_LEADER
— VM_CF_SIGNAL · VM_LANE_CHANGE
```

- ▶ **You derive, TransModeler calls** — no polling, no marshalling, data arrives *by value*.
- ▶ **Opt in per vehicle** — the VM_* flags you set in AttachVehicle decide which callbacks fire and what they carry. *Cost is what you use*.
- ▶ **Return flt_miss to defer** — TransModeler keeps its own value for that step.
- ▶ **Safety still enforced** — your acceleration is clamped; a plugin cannot run a red light or drive through its leader.
- ▶ **Fully multi-threaded** — callbacks fire on worker threads, so plugin state must be thread-safe.

C. Vehicle Monitor — signal data delivered to the driver

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Exploration

```
struct SVehicleNextSignal {
    LPCSTR rsuid;           // RSU it came from
    float fDistance;        // m to the signal
    float fTravelTime;      // s to the signal
    long idNode;
    long idSignal;
    TsmTurn iTurn;          // movement
    TsmSignalState iState;  // CURRENT state
    float fTimeOn;          // state began
    float fTimeOffMin;      // earliest change
    float fTimeOffMax;      // latest change
};
```

Delivered two ways: the NextSignal() callback, and — under VM_CF_SIGNAL — *embedded in SCarFollowingData*, so signal timing arrives in the same call where acceleration is decided.

Against J2735 TimeChangeDetails:

startTime	fTimeOn
minEndTime	fTimeOffMin
maxEndTime	fTimeOffMax
likelyTime	absent
confidence	absent

For pretimed controllers all three return the same value. For traffic actuated control, SignalTimeOff() returns the average of min/max values.

Why min/max matters — under actuation the end of green is a *range*. GLOSA on a point estimate flatters itself in simulation, disappoints in the field.

Scope — per *vehicle*, its next signal; J2735 SPaT is per *intersection*, all movements. SPaT semantics, not a SPaT message.

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Demo

RSU emulation — imperfection as a modeling primitive

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A **point** RSU is a centre and a radius; an **area** RSU is a polygon.

Footprint, not pattern. The point case is the idealized *omnidirectional* coverage; a polygon can *represent* a directional or sector footprint without being parameterized as one — no boresight, beamwidth, gain, or elevation pattern.

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Demo

Knob	What it models	Status
Active	hardware or software malfunction — toggled <i>during</i> a run, per RSU	works
RndSpatial	standard deviation of relative random error to shorten the communication range — drawn per RSU, clamped to e.g. $\pm 25\%$, applied to radius and polygon	works
RndTemporal	binary probability of random error to skip the scheduled communication	inert

Being precise about RndTemporal — the per-RSU probability is drawn and clamped, then *read nowhere*. No message currently traverses an RSU on its way to a vehicle, so there is nothing to drop. Packet loss today is done in user code, at the hook on the next slide.

The BSM we emulate

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```
typedef struct STsmBSM {
    LONG    idTemporary;    // rotating ID
    LONG    lMessageCount;
    DOUBLE  dUtcTimestamp;  // UTC seconds
    LONG    lLongitude;     // 1e-6 degrees
    LONG    lLatitude;      // 1e-6 degrees
    FLOAT   fElevation;     // m
    TsmTransmissionStateType tState;
    FLOAT   fSpeed;         // m/s
    SHORT   fHeading;       // azimuth, deg
    FLOAT   fAcceleration;  // m/s2, long.
    FLOAT   fLength, fWidth; // m
    BYTE    iClass;         // vehicle class
} STsmBSM;
```

Pseudonym rotation — idTemporary is re-randomized every BSMIDScramble messages, so a vehicle is not trackable by identifier alone.

Against J2735 BSM Part I:

msgCnt	lMessageCount
id	idTemporary
secMark	dUtcTimestamp
lat/long/elev	lLatitude/...
transmission	tState
speed	fSpeed
heading	fHeading
size	fLength/fWidth
accelSet (4 axes)	longitudinal only
accuracy	absent
angle (steering)	absent
brakes	absent

Same units as the standard — 10^{-6} degrees, m/s, degrees azimuth.

A struct, not an encoding — no ASN.1/UPER, no envelope, no signature. The simulator does not own wire formats — a user application injects the encoding for the system of interest.

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Demo

The hook: BSMGenerate

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Microsimulation API for V2X
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```
// _IVehicleEvents -- enable with
// EnableVehicleCallback(
//     VCB_BSMGENERATE, this, true)
```

```
HRESULT BSMGenerate(
    [in]      DOUBLE    dTime,
    [in]      ITsmVehicle* pVehicle,
    [in, out] STsmBSM* pBSM,
    [out, retval] DOUBLE* pNextTime);
```

Fires **before any other vehicle or RSU reads the message.**

Whatever you write into pBSM *persists* — it is exactly what

ITsmVehicle::BSM returns to everyone else.

What it makes researchable

- ▶ **error** — perturb position, speed, heading
- ▶ **spoofing** — write values unrelated to truth
- ▶ **loss** — restore the previous BSM, so the receiver's picture goes stale
- ▶ **latency** — hold back dUtcTimestamp
- ▶ **silent period** — suppress broadcast after a pseudonym change
- ▶ **OBU failure** — retire the vehicle entirely

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The TransModeler V2X API
Demo

Demo

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Microsimulation API for V2X
Exploration

► GUI Demo

► Code Demo

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[Demo](#)

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Exploration

Questions?

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microsimulator API
The TransModeler V2X API
[Demo](#)